BOUNDED COMPLEXES WITH PERFECT COHOMOLOGY MODULES ARE PERFECT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Two-out-of-three for perfect objects: in a distinguished triangle, if two terms are perfect then so is the third. invoked at proof lines 18994-18996. Closes the induction step by passing from the truncated complex and the last cohomology module to the full complex.

Source reference. Original lemma 066R: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18949–18949. Cited at proof lines 18994-18996 to close the induction step; the two-out-of-three property is load-bearing inside the counted span.

Lemma 0.2. *Let R be a ring. If $K^\bullet \in D^b(R)$ and all its cohomology modules are perfect, then $K^\bullet$ is perfect.*

Proof. Follows by induction on the length of the finite complex: use Lemma 066R and the canonical truncations. □

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